

pharmaniaga®

Bupivacaine Hydrochloride 0.5% w/v Injection



COMPOSITION
Each 20mL contains Bupivacaine Hydrochloride (anhydrous) 100mg.

DESCRIPTION
Clear or almost colourless solution

PHARMACODYNAMICS
Bupivacaine is a long acting, amide-type local anaesthetic with both anaesthetic and analgesic effects. At high doses it produces surgical anaesthesia, while at lower doses it produces sensory block (analgesia) with less pronounced motor block. Onset and duration of the local anaesthetic effect of bupivacaine depends on the dose and site of administration.

Bupivacaine like other local anaesthetics, causes a reversible blockade of impulse propagation along nerve fibres by preventing the inward movement of sodium ions through the nerve membrane. The sodium channel of the nerve membrane is considered a receptor for local anaesthetic molecules.

Local anaesthetics may have similar effects on other excitable membranes, e.g. in the brain and myocardium. If excessive amounts of medicine reach the systemic circulation, symptoms and signs of toxicity may appear, emanating mainly from the central nervous and cardiovascular systems.

Central nervous system toxicity usually precedes the cardiovascular effects as it occurs at lower plasma concentrations. Direct effects of local anaesthetics on the heart include slow conduction, negative inotropism and eventually cardiac arrest.

Indirect cardiovascular effects (hypotension, bradycardia) may occur after epidural administration or spinal administration depending on the extent of the concomitant sympathetic block.

In concentrations of 5 mg/ml, bupivacaine has a long duration of action, from 2 to 5 hours following a single epidural injection and up to 12 hours after peripheral nerve blocks. The onset of blockade is slower than with lignocaine, especially when anaesthetising large nerves. When used in low concentrations (2.5 mg/mL or less) there is less effect on motor nerve fibres and the duration of action is shorter. Low concentrations may, however, be used with advantage for prolonged pain relief, e.g. in labour or postoperatively.

PHARMACOKINETICS
Absorption
The plasma concentration of bupivacaine depends upon the dose, the route of administration and the vascularity of the injection site.

After injection of bupivacaine solutions for caudal, epidural or peripheral nerve block in man, peak plasma levels of bupivacaine in the blood are attained within 30 to 45 minutes, followed by a decline to insignificant levels during the next 3 to 6 hours.

Intercostal blocks give the highest peak plasma concentration due to a rapid absorption (maximum plasma concentrations in the order of 1 to 4 mg/L after a 400 mg dose), while subcutaneous abdominal injections give the lowest plasma concentration. Epidural and major plexus blocks are intermediate. In children rapid absorption and plasma concentrations (in the order of 1 to 1.5 mg/L after a dose of 3 mg/kg) is seen with caudal block.

Absorption of bupivacaine from the epidural space occurs in 2 phases; the first phase is in the order of 7 minutes and the second is in 6 hours. The slow absorption is rate-limiting in the elimination of bupivacaine, which explains why the apparent elimination half-life after epidural administration is longer than after intravenous administration.

Distribution
Bupivacaine has a volume of distribution at steady state of 73 litres after IV administration. It is mainly bound to α 1-acid glycoprotein in plasma with a plasma binding of 96%.
An increase in α 1-acid glycoprotein, which occurs postoperatively after major surgery, may cause an increase in the total plasma concentration of bupivacaine. The level of free drug will remain the same. This explains why total plasma concentrations above the apparent toxic threshold level of 2.6 to 3.0 mg/L are apparently well tolerated in this situation.

Elimination
Bupivacaine has a total plasma clearance of 0.58 L/min, an elimination half-life of 2.7 hours and an intermediate hepatic extraction ratio of 0.40 after IV administration. The terminal elimination half-life is prolonged in the newborn to approximately 8 hours. In children aged over 3 months the elimination half-life is similar to that in adults.

Bupivacaine is excreted in the urine principally as metabolites with about 6% as unchanged drug. Following epidural administration, the urinary recovery of unchanged bupivacaine is about 0.2%, of pipercolylxylinide (PPX) about 1% and of 4-hydroxy-bupivacaine about 0.1% of the administered dose.

Special patient considerations

Various pharmacokinetic parameters can be significantly altered by a number of factors including the presence of hepatic and renal disease, route of administration, age of the patient and certain concomitant medication.

INDICATION
Bupivacaine Hydrochloride 0.5% w/v Injection are indicated for the production of local or regional anaesthesia and analgesia in individuals as follows:-

Surgical anaesthesia

- Epidural block for surgery
- Field block (minor and major nerve blocks and infiltration)

Analgesia

- Continuous epidural infusion or intermittent bolus epidural administration for analgesia in postoperative pain or labour pain
- Field block (minor nerve block and infiltration)

RECOMMENDED DOSAGE
As with all local anaesthetics, the dosage varies and depends upon the area to be anaesthetised, the vascularity of the tissues, the number of neuronal segments to be blocked, the depth of anaesthesia and degree of muscle relaxation required, individual tolerance, the technique of anaesthesia, and the physical condition of the patient.

The lowest dosage that results in effective anaesthesia should be used. In general, surgical anaesthesia requires the use of higher concentrations and doses than those required for analgesia. The volume of the drug used will affect the extent of spread of anaesthesia.

The presentations of Bupivacaine Injection are intended for single use only. Any solution remaining from an opened container should be discarded.

The following tables are a guide to dosage. The clinician's experience and knowledge of the patient's physical status are of importance in deciding the dose. Experience to date indicates that 400 mg administered over 24 hours is well tolerated in average adults.

Recommended dosage for bupivacaine solutions for various anaesthetic procedures in the average, healthy 70 kg adult patient are as follows:

RECOMMENDED DOSAGE FOR SURGICAL ANAESTHESIA			
SURGICAL ANAESTHESIA	Conc. mg/mL	DOSE	
		mL	Mg
Lumbar Epidural			
Abdominal, pelvic and lower limb surgery, including Caesarean section	5.0	15 to 30	75 to 150
Thoracic Epidural			
Upper Abdominal and thoracic surgery	2.5	5 to 15	12.5 to 37.5
	5.0	5 to 10	25 to 50
Caudal Epidural	2.5	15 to 40	37.5 to 100
	5.0	15 to 25	75 to 125
Other blocks			
Local infiltration	2.5	5 to 60	12.5 to 150
	5.0	5 to 30	25 to 150
Intercostal (per segment)	2.5	4 to 8	10 to 20
	5.0	3 to 5	15 to 25
Brachial plexus	5.0	20 to 30	100 to 150
Sciatic	5.0	10 to 20	50 to 100
3 in 1 (Femoral, obturator and lateral cutaneous)	5.0	10 to 20	50 to 100
Pudendal	2.5 to 5.0	5 to 10 each side	7.5 to 100

RECOMMENDED DOSAGE FOR ANAESTHESIA, INCLUDING CONTINUOUS INFUSION

ANAESTHESIA	Conc. mg/mL	DOSE	
Caudal Epidural			
Postoperative pain management	2.5	20 to 30 mL bolus	50 to 75 mg bolus
Lumbar Epidural			
Bolus (incl. labour pain management)	2.5 to 5.0	6 to 12 mL bolus followed by	15 to 60 mg bolus followed by
Continuous infusion (incl. labour pain and postoperative pain management)	1.25*	10 to 15mL/hour	12.5 to 18.75 mg/hour
	2.5	5 to 7.5 mL/hour	12.5 to 18.75 mg/hour
Thoracic Epidural			
Continuous infusion for postoperative pain management	1.25*	5 to 10mL/hour	6.25 to 12.5 mg/hour

*This solution is mainly used for epidural administration in combination with a suitable opioid for postoperative pain management. For further details of procedures please see current, standard textbooks.

NOTE:

1. Recommended doses: tolerability varies widely between patients and toxic effects may occur after any local anaesthetic procedure. Careful observation of the patient must therefore be maintained. It is recommended that the dose of bupivacaine at any time should not exceed 2 mg/kg. However, the dose administered must be tailored to the individual patient and procedure, and the maximum dose quoted here should be used as a guide only.

2. Injection: injection of repeated doses of bupivacaine may cause significant increase in blood levels with each repeated dose, due to accumulation of the drug or its metabolites, or due to slow metabolic degradation.

The rapid injection of a large volume of local anaesthetic solution should be avoided and fractional doses should be used when feasible. For most indications the duration of bupivacaine is such that a single dose is sufficient.

3. Hypotension: during thoracic, lumbar and caudal epidural anaesthesia/analgesia, a marked fall in blood pressure and/or intercostal paralysis may be seen, possibly due to the use of excessive doses, improper positioning of the patient or accidental disposition of the anaesthetic within the subarachnoid space. Hypotension and bradycardia may occur as a result of sympathetic blockade.

4. Test dose: for epidural anaesthesia, a 3 to 5 mL test dose of local anaesthetic solution, preferably containing up to 15 micrograms adrenaline, should be administered. Verbal contact and repeated monitoring of the heart rate and blood pressure should be maintained for 5 minutes following the test dose after which, in the absence of signs of subarachnoid or intravascular injection, the main dose may be given.

Use of a test dose containing adrenaline may have further advantages in that an intravascular injection of adrenaline will be quickly recognised by an increase in heart rate, usually within about 40 seconds. To detect this, the heart rate and rhythm should be monitored with an electrocardiogram.

An accidental intrathecal injection may be recognised by signs of a spinal block.

Prior to administration of the total dose, aspiration should be repeated. The main dose should be injected slowly at a rate of 25 to 50 mg/min, while closely observing the patient's vital functions and maintaining verbal contact. If toxic symptoms or signs occur, the injection should be stopped immediately.

5. Prolonged blocks: When prolonged blocks are used, either by continuous infusion or by repeated bolus administration, the risks of reaching a toxic plasma concentration or inducing a local neural injury must be considered.

Use in Children

Experience with bupivacaine in children under the age of 12 is limited. The dosage in children should be calculated on a weight basis up to 2 mg/kg.

Use in Pregnancy

It should be noted that the dose should be reduced in patients in the late stages of pregnancy.

Use in Debilitated or Elderly Patients

Debilitated or elderly patients, including those with partial or complete heart block, advanced liver disease or severe renal dysfunction should be given a reduced dosage commensurate with their physical condition.

ROUTE OF ADMINISTRATION

Parenteral : For field block and epidural administration only

CONTRAINDICATIONS

- Allergy or hypersensitivity to amide type local anaesthetics. Detection of suspected hypersensitivity by skin testing is of limited value.
- Epidural and spinal anaesthesia is contraindicated in patients with uncorrected hypotension.
- Local anaesthetics should not be used when there is infection in the region of the proposed injection and/or in the presence of septicaemia.
- Bupivacaine is contraindicated in obstetric paracervical block, intravenous regional anaesthesia (Bier's block) and all intravenous infusion
- General contraindications related to epidural anaesthesia, regardless of the local anaesthetic used, should be taken into account.
- Epidural anaesthesia, regardless of the local anaesthetic used, has its own contraindications which include: active disease of the central nervous system such as meningitis, poliomyelitis, intracranial haemorrhage, subacute combined degeneration of the cord due to pernicious anaemia, and cerebral or spinal tumours; tuberculosis of the spine; pyogenic infection of the skin at or adjacent to the site of lumbar puncture; cardiogenic or hypovolaemic shock; coagulation disorders or ongoing anticoagulant treatment.
- Epidural and spinal anaesthesia is contraindicated in patients with an expanding cerebral lesion, a tumour, cyst or abscess, which may, if the intracranial pressure is suddenly altered, cause obstruction to the cerebrospinal fluid or blood circulation (the pressure cone).

WARNINGS AND PRECAUTIONS

During administration of bupivacaine, it is mandatory to have emergency resuscitative equipment (including oxygen) and drugs immediately available to manage possible adverse reactions involving the cardiovascular, respiratory or central nervous systems.

Because of the possibility of hypotension and bradycardia following major blocks, an IV cannula should be inserted before the local anaesthetic is injected. Delay in proper management of dose-related toxicity, under-ventilation from any cause and/or altered sensitivity may lead to the development of acidosis, cardiac arrest and death.

The safety and effectiveness of bupivacaine depend upon proper dosage, correct technique and adequate precautions. Standard textbooks should be consulted for specific techniques and precautions for various regional anaesthetic procedures.

- Injections should always be made slowly with frequent aspirations to avoid inadvertent intravascular injection which can produce toxic effects.
- Careful and constant monitoring of cardiovascular and respiratory vital signs and the patient's state of consciousness should be accomplished after each local anaesthetic injection. It should be kept in mind that at such times restlessness, anxiety, tinnitus, dizziness, blurred vision, tremors, depression or drowsiness may be early warning signs of CNS toxicity.
- The use of local anaesthetics for major peripheral nerve block may involve

the administration of large volumes in highly vascularised areas, often close to large blood vessels. As such there is an increased risk of intravascular injection and/or systemic absorption which can lead to high plasma concentrations. There have been reports of cardiac arrest or death during the use of bupivacaine for epidural anaesthesia or peripheral nerve blockade. In some instances, resuscitation has been difficult or impossible despite apparently adequate preparation and management.

- Central nerve blocks may cause cardiovascular depression, especially in the presence of hypovolaemia. Epidural anaesthesia should be used with caution in patients with impaired cardiovascular function. Epidural anaesthesia may lead to hypotension and bradycardia.

- Low Molecular Weight Heparins and Heparinoids (Spinal/Epidural Haematomas) – When neuraxial anaesthesia (epidural/spinal anaesthesia) is employed, patients anti-coagulated or scheduled to be anti-coagulated with low molecular weight heparins or heparinoids are at risk of developing an epidural or spinal haematoma which can result in long-term or permanent paralysis. The risk of these events is increased by the use of indwelling epidural catheters, traumatic or repeated epidural/spinal puncture, and the concomitant use of drugs affecting haemostasis such as non-steroidal anti-inflammatory drugs (NSAIDs), platelet inhibitors or other anticoagulants. Patients should be frequently monitored for signs and symptoms of neurological impairment.

- The lowest dosage that results in effective anaesthesia should be used. Repeated injection of Bupivacaine Injection may cause accumulation of bupivacaine or its metabolites and result in toxic effects.

Tolerance to elevated blood levels varies with the status of the patient. Elderly, young or debilitated patients, including those with partial or complete conduction block, advanced liver disease or severe renal impairment, should be given reduced doses commensurate with their age and physical condition.

Caution should be used when administering bupivacaine to children under 12 years of age.

- Bupivacaine may cause acute toxicity effects on the central nervous and cardiovascular systems if utilised for local anaesthetic procedures resulting in high blood concentrations of the drug. This is especially the case after unintentional intravascular administration. Ventricular arrhythmia, ventricular fibrillation, sudden cardiovascular collapse and death have been reported in connection with high systemic concentrations of bupivacaine.

- Bupivacaine should be given with caution to patients with epilepsy, impaired cardiac conduction, bradycardia, severe shock or digitalis intoxication. It should also be administered with caution to patients with impaired cardiovascular function as they may be less able to compensate for functional changes associated with the prolongation of AV conduction produced by bupivacaine. Patients being treated with anti-arrhythmic drugs class III (e.g. amiodarone) should be under close surveillance and ECG monitoring since cardiac effects may be additive.

- In patients with Stokes-Adams syndrome or Wolff-Parkinson-White syndrome extreme care should be taken to avoid accidental arterio-venous injection.

- Local anaesthetics should be given with great caution (if at all) to patients with pre-existing neurological or neuromuscular disease e.g. myasthenia gravis. Use with extreme caution in epidural, caudal and spinal anaesthesia when there are serious diseases of the CNS or of the spinal cord, e.g. meningitis, spinal fluid block, cranial or spinal haemorrhage, tumours, poliomyelitis, syphilis, tuberculosis or metastatic lesions of the spinal cord.

- Bupivacaine is eliminated primarily by hepatic metabolism and changes in hepatic function may have significant consequences. Bupivacaine has an intermediate clearance which depends on its unbound fraction and intrinsic metabolic clearance. Bupivacaine should therefore be used with caution in patients with severe hepatic disease.

- Bupivacaine should be used with caution in patients with severe renal dysfunction because acidosis and reduced plasma protein concentration, which are frequently seen in these patients, may increase the risk of systemic toxicity.

- Patients with hyperthyroidism are also more susceptible to toxicity with bupivacaine.

- Inadvertent intravascular or subarachnoid injection of small doses of local anaesthetics injected into the head and neck area, including retrobulbar, dental and stellate ganglion blocks may produce adverse reactions similar to systemic toxicity seen with unintentional intravascular injections of larger doses. Injections made inadvertently into an artery may cause immediate cerebral symptoms even at low doses.

Clinicians who perform retrobulbar blocks should be aware that there have been reports of cardiovascular collapse and apnoea following the

use of local anaesthetic injections for retrobulbar block. Prior to retrobulbar block, necessary equipment, drugs and personnel should be immediately available as with all other regional procedures. Retrobulbar injections may very occasionally reach the subarachnoid space, causing temporary blindness, cardiovascular collapse, apnoea, convulsions etc. These must be diagnosed and treated promptly.

Retro- and peribulbar injections of local anaesthetics carry a low risk of persistent ocular muscle dysfunction. The primary causes include trauma and /or local toxic effects on muscles or nerves. The severity of such tissue reactions is related to the degree of the trauma, the concentration of the local anaesthetic and the duration of exposure of the tissue to the local anaesthetic. For this reason, as with all local anaesthetics, the lowest effective concentration and dose of local anaesthetic should be used. Vasoconstrictors may aggravate tissue reactions and should be used only when indicated.

- Fetal bradycardia/tachycardia frequently follows paracervical block with some amide type local anaesthetics and may be associated with fetal acidosis and hypoxia. Added risk appears to be present in prematurity, toxemia of pregnancy and fetal distress. Careful monitoring of the fetal heart rate is necessary.

- Bupivacaine should be used with caution in patients with known drug sensitivities.

Local anaesthetics react with certain metals and cause the release of their respective ions which, if injected, may cause severe local irritation. Adequate precautions should be taken to avoid prolonged contact between bupivacaine solutions and metal surfaces such as metal bowls, cannulae and syringes with metal parts.

INTERACTION WITH OTHER MEDICAMENTS

Bupivacaine should be used with caution in patients receiving other local anaesthetics or agents structurally related to amide-type local anaesthetics, e.g. certain anti-arrhythmic drugs such as mexiletine and lignocaine, since potentiation of cardiac effects may occur.

Specific interaction studies with Bupivacaine and anti-arrhythmic drugs class III (e.g. amiodarone) have not been performed, but caution should be advised.

Incompatibilities

Bupivacaine Injection should not be mixed with other drugs. The solution must not be stored in contact with metals, e.g. needles or metal parts of syringes, as dissolved metal ions may cause swelling at the site of injection.

PREGNANCY AND LACTATION

Use in Pregnancy - Category A

After epidural administration of bupivacaine to women in labour, bupivacaine crosses the placental barrier. However, concentrations in umbilical veins are lower than those found in the maternal circulation.

Bupivacaine has been effectively used for obstetrical analgesia and adverse effects on the course of labour or delivery are rare. It has been suggested that blood glucose levels should be checked in newborns after obstetric regional anaesthesia.

Fetal adverse effects due to bupivacaine, such as fetal bradycardia, seem to be most apparent in paracervical block anaesthesia. Such effects may be due to high concentrations of anaesthetic reaching the fetus.

The safe use of bupivacaine during pregnancy, other than labour, has not been established. Although bupivacaine has been used extensively for surgical procedures during pregnancy with no reports of ill effects to mother or fetus, there are no adequate and well-controlled studies in pregnant women of the effect of bupivacaine on the developing fetus. It is known that local anaesthetic cross the placenta rapidly. It should therefore be used cautiously during pregnancy other than labour.

Use in Lactation

Bupivacaine passes into breast milk. The amount of bupivacaine appearing in breast milk from a breastfeeding mother receiving parenteral bupivacaine is unlikely to lead to a significant accumulation of the parent drug in the breast-fed infant.

At maternal serum levels of up to 0.45 µg/mL produced by the epidural use of bupivacaine for vaginal delivery, bupivacaine could not be detected in breast milk during the first 24 hours after delivery (detection limit 0.02 µg/mL).

The possibility of an idiosyncratic or allergic reaction in the breast-fed infant from bupivacaine remains to be determined.

ADVERSE EFFECT

The adverse reactions profiles for bupivacaine is similar to those observed with other amide-type local anaesthetics and pertain mainly to the central nervous system and the cardiovascular system. Adverse reactions to bupivacaine are, in general, dose-related and may result from high plasma levels caused by excessive dosage, rapid absorption, delayed elimination, altered metabolism, inadvertent intravascular injection or may result from a

hypersensitivity, idiosyncrasy or diminished tolerance on the part of the patient.

Serious adverse experiences are generally systemic in nature. Ventricular arrhythmias, ventricular fibrillation, sudden cardiovascular collapse and death have been reported when bupivacaine has been utilised for local anaesthetic procedures that may result in high systemic concentrations of bupivacaine.

Pronounced acidosis, hyperkalaemia, hypocalcaemia or hypoxia in the patient may increase the risk and severity of toxic reactions.

Central Nervous System

CNS manifestations are excitatory and/or depressant and may be characterised by light-headedness, nervousness, apprehension, euphoria, confusion, dizziness, drowsiness, tinnitus, blurred vision, diplopia, nausea, vomiting, sensations of heat, cold or numbness, urinary retention, paraesthesia circumoral, paraesthesia, hyperacusis, twitching, tremors, convulsions, unconsciousness, respiratory depression and/or arrest, agitation, numbness of the tongue, difficulty swallowing and slurred speech.

The excitatory manifestations may be very brief or may not occur at all, in which case the first manifestation of toxicity may be drowsiness merging into unconsciousness and respiratory arrest. Drowsiness following administration of bupivacaine is usually an early sign of a high blood level of the drug and may occur as a result of rapid absorption. In unconscious patients, circulatory collapse should be watched as CNS effects may not be apparent as an early manifestation of toxicity may in some cases progress to frank convulsions and ultimately lead to respiratory depression and/or arrest.

Cardiovascular

Cardiovascular manifestations following inadvertent intravascular injection are usually depressant and are characterised by bradycardia, hypotension, and cardiovascular collapse, which may lead to cardiac arrest.

Haemodynamic

Regional anaesthesia may lead to maternal hypotension.

Neurologic

The incidences of adverse reactions associated with the use of local anaesthetics may be related to the total dose of local anaesthetic administered and are also dependent on the particular drug used, the route of administration and the physical status of the patient.

In the practice of caudal or lumbar epidural block, occasional unintentional penetration of the subarachnoid space by the catheter may occur. Subsequent adverse effects may depend partially on the amount of drug administered subdurally.

These may include spinal block of varying magnitude (including total spinal block), hypotension secondary to spinal block, loss of bladder and bowel control and loss of perineal sensation and sexual function. Persistent motor, sensory and/or autonomic (sphincter control) deficit of some lower spinal segments with slow recovery (several months) or incomplete recovery have been reported in rare instances when caudal or lumbar epidural block has been attempted. Backache and headache have also been noted following use of these anaesthetic procedures.

Paresis, paraplegia, neuropathy, peripheral nerve injury and arachnoiditis have been observed.

The effects of systemic overdose and unintentional intravascular injection may involve the central nervous system and/or the cardiovascular system. Inadvertent subarachnoid injection may lead to CNS depression, respiratory arrest and cardiovascular collapse.

Allergic

Allergy to amide-type local anaesthetics is rare. These allergic reactions are characterised by signs such as cutaneous lesions, urticaria, oedema or anaphylactoid reactions. If such a reaction occurs, it should be managed by conventional means. The detection of sensitivity by skin testing is of doubtful value.

Musculoskeletal

Although intra-articular continuous infusions of local anaesthetics following arthroscopic and other surgical procedures is an unapproved use, there have been post-marketing reports of chondrolysis in patients receiving such infusions. Symptoms include joint pain, stiffness and loss of motion.

Currently, there is no effective treatment for chondrolysis. Patients who experienced chondrolysis have required additional diagnostic and therapeutic procedures and some required arthroplasty or shoulder replacement. Therefore, bupivacaine should not be used for post-operative intra-articular continuous infusion.

SYMPTOM & TREATMENT OF OVERDOSE

Symptoms of Acute Toxicity

Acute emergencies associated with the use of local anaesthetics are generally related to high plasma levels or to unintended subarachnoid

injection of the local anaesthetic solution.

With accidental intravascular injections of local anaesthetics, the toxic effects will be obvious within 1 to 3 minutes. With overdose, peak plasma concentrations may not be reached for 20 to 30 minutes, depending on the site of injection and toxic signs will be delayed. Toxic reactions mainly involve the central nervous and cardiovascular systems.

Central nervous system toxicity is a graded response with symptoms and signs of escalating severity. The first symptoms are circumoral paraesthesia, numbness of the tongue, lightheadedness, hyperacusis and tinnitus. Visual disturbances and muscular tremors are more serious and precede the onset of generalised convulsions. These signs must not be mistaken for neurotic behaviour.

Unconsciousness and grand mal convulsions may follow. These may last from a few seconds to several minutes. Hypoxia and hypercapnia occur rapidly following convulsions due to increased muscular activity, together with the interference with normal respiration and loss of the airway. In severe cases, apnoea may occur. Acidosis, hyperkalaemia, hypocalcaemia and hypoxia increase and extend the toxic effects of local anaesthetics.

Recovery is due to redistribution of the local anaesthetic drug from the central nervous system and metabolism. Recovery may be rapid unless large amounts of the drug have been injected.

Signs of Cardiovascular toxicity indicates a more severe situation. Hypotension, bradycardia, decreased cardiac output, heart block, arrhythmia and even ventricular arrhythmias, ventricular fibrillation and cardiac arrest may occur as a result of high systemic concentrations of local anaesthetics.

Cardiovascular toxic effects are generally preceded by signs of toxicity in the central nervous system, unless the patient is receiving a general anaesthetic or is heavily sedated with drugs such as benzodiazepines or barbiturates.

Treatment of Overdosage

If signs of acute systemic toxicity appear, injection of the local anaesthetic should be immediately stopped.

If convulsions occur then immediate attention is required for the maintenance of patent airway and assisted or controlled ventilation with oxygen, via a positive airway pressure delivery system mask. Adequacy of the circulation should then be evaluated, bearing in mind that drugs used to treat convulsions depress the circulation when administered intravenously.

Should convulsions persist despite adequate respiratory support, and if the status of the circulation permits, appropriate anticonvulsant medication such as an ultra-short acting barbiturate (e.g. thiopentone) or a benzodiazepine (e.g. diazepam) may be administered IV. The clinician should be familiar with these anticonvulsant drugs prior to use of local anaesthetics.

Suxamethonium will stop the muscle convulsions rapidly but will require tracheal intubation and controlled ventilation, and should only be used by those familiar with these procedures.

If ventricular fibrillation or cardiac arrest occurs, effective cardiovascular resuscitation treatment must be instituted and maintained for a prolonged period if necessary. Optimal oxygenation and ventilation, and circulatory support as well as treatment of acidosis are of vital importance.

EFFECTS ON ABILITY TO DRIVE AND USE OF MACHINES

Depending on the dosage, local anaesthetics may have a mild effect on mental function and coordination and may temporarily impair locomotion and coordination.

INSTRUCTIONS FOR USE

If only part used, discard the remaining solution.

STORAGE CONDITION

Store below 30°C.
Retain in carton till time of use.

SHELF LIFE

3 years from date of manufacture.

Once opened discard any unused portion.

DOSAGE FORM AND PACKAGING AVAILABLE

5 x 20mL vial (clear glass vial).
5 vials per unit PVC tray were packed in paper carton.

PRODUCT REGISTRATION HOLDER/ MANUFACTURER PHARMANIAGA LIFESCIENCE SDN BHD (198201002939)

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2003748-V

Revision Date: 21-07-2022